

A Taxonomy for Guiding XR Prototyping Decisions by the Non-Tech-Savvy

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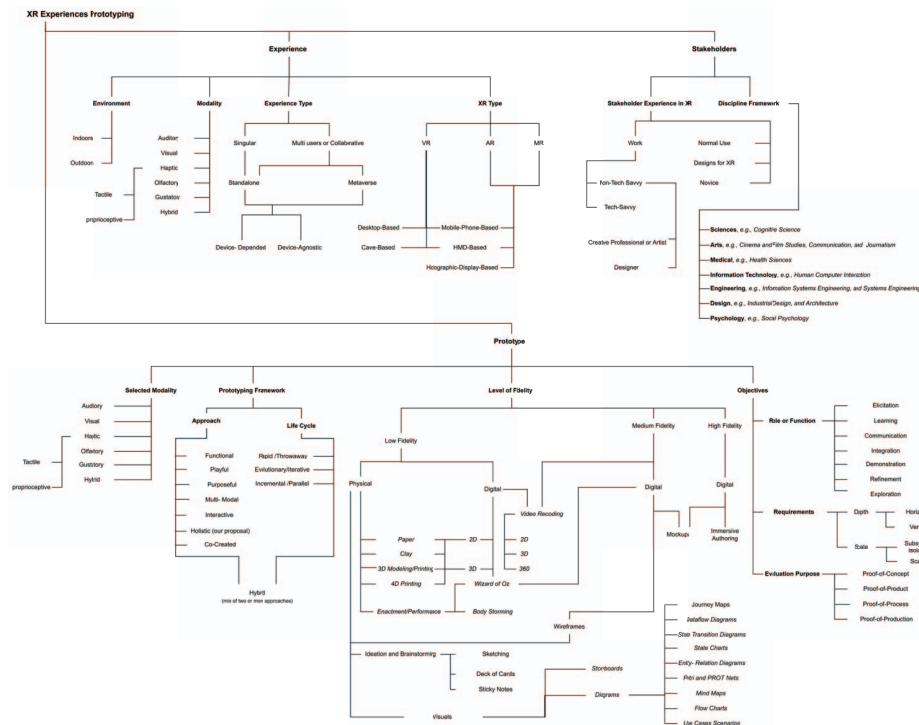


Figure 1: Our Proposed Holistic XR Experiences Prototyping Taxonomy

ABSTRACT

Within XR design, prototypes play an essential role in materializing, communicating and evaluating concepts. Yet, selecting the appropriate prototype proves intuitive for seasoned designers but daunting for beginners and the non tech-savvy. Addressing the right challenges when making decisions on the prototyping method is vital for a constructive outcome. Taxonomies emerge as powerful tools in this context. We forge a holistic taxonomy based on various prototyping taxonomies in the contexts of XR and other disciplines. We further validate it by surveying existing XR prototyping work. Next, we will further validate it through a series of steps common in such research.

Index Terms: Human-centered computing—Interaction design—Interaction design process and methods—Interface design prototyping; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Extended reality (XR) shows great potential as a storytelling medium [6]. This potential has increased interest in XR design from across

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both novice and non-tech-savvy researchers, designers, and artists. In most cases, these applications could be forgiving, however, some issues related to this medium could hinder the growth of such applications due to the way they affect experiences.

Designing XR experiences poses significant challenges for novice non tech-savvy designers. In the interdisciplinary area of XR, they often struggle with their unfamiliarity with XR technologies, limited knowledge of design principles specific to XR [1, 7], and the complexity of integrating technical aspects with creative content [1, 7]. Due to their limited knowledge, they often face difficulties in understanding the specific needs and preferences of XR users, and the issues presented by these media, like guiding attention and cyber-sickness, leading to sub-optimal user experiences.

The findings of design research tend to be difficult to implement or locate and in some cases abstract or incomprehensible [7]. There is a lack of clear design guidelines when it comes to XR development, which makes each designer prone to bias based on their knowledge and research ecosystem [1]. As a result, it is crucial to develop translational resources [3] that can bridge the gap by removing these obstacles and easing the process. XR prototyping is an area we seriously lack in having enough tools to support the process [1]. Prototyping allows designers to test and refine their ideas, identify potential challenges, and gather feedback from users before moving forward with full-scale development of XR experiences. Therefore, it is an important area to focus on as a start. Furthermore, XR design process requires tools that support design and prototyping decision making. However, there is a lack of holistic processes and decision

making tools for prototyping in XR development. We propose this taxonomy not only as a classification tool but also as a decision making tool that could aid novices.

To address these issues, we synthesize insights from various comprehensive prototyping taxonomies in the contexts of XR and other disciplines to forge a holistic decision-making guide. We further validate our taxonomy by surveying existing XR prototyping work in the literature. Intended primarily for novices and those non tech-savvy, we hypothesize that our taxonomy also has the potential of becoming a valuable educational resource to introduce learners to XR prototyping nuances.

2 PROPOSED XR EXPERIENCE PROTOTYPING TAXONOMY

There is a significant body of research when it comes to identifying XR related prototyping methods and building tools and artifacts that facilitate the process. As a start, we surveyed 50 of the main ones as a quick overview utilizing a search method similar to that of Freitas et al. [5]. Given the long history of prototyping and the rich literature in different fields, it might not be possible to create a complete and exhaustive list as there are potentially hundreds of prototyping techniques and artifacts, each with unique variations. Therefore, we focus on prototyping taxonomies and techniques which are valid within the context of prototyping for XR experiences. This will highlight some of the lessons learned from other disciplines. The logic behind examining other disciplines is that they have been developing this process, i.e., prototyping model, over decades, and the practice has reached a level of maturity, especially in the area of prototyping taxonomies. In the rapidly evolving interdisciplinary field of XR, we should take a closer look at prototypes taxonomies to establish a universal language and decision making process regardless of the stakeholders' disciplines or frameworks. We hypothesize that such an effort would help in facilitating better design education within the context of XR Design, better working conditions for non-tech-savvy designers and artists, and faster transformation of this field to become transdisciplinary, in a manner similar to how film developed as an independent art form.

To produce our proposed taxonomy for XR Experience Prototyping, we evaluated 13 prominent taxonomies in HCI and different design spaces, between the years (1984–2019). The interest in prototyping taxonomies has been more pronounced in engineering and product design disciplines with some modest attempts in HCI, very few of which are within XR contexts. Therefore, the majority of these mature taxonomies are inherited from engineering and product design frameworks, yet, they are very applicable and informative within the interdisciplinary context of XR experiences prototyping. By examining these taxonomies, we notice that they classify prototypes based on different classes, each of which branches into characteristics that could mean different things in different context. However, many of them are applicable within the context of prototyping XR experiences after adding some unique classes that apply to the medium solely.

Our proposed taxonomy (see Figure 1) is a result of our literature review, evaluating the aforementioned taxonomies, reflecting on them and their rationale, and identifying XR Experiences related characteristics and factors. This taxonomy classifies XR experiences prototypes based on three main classifiers; these are: the desired experience, the stakeholders, and the prototype. Then these are broken down into sub-classifiers.

First, the Experience consists of (i) Environment, (ii) Modality, these modalities are based on the senses and could be visual, auditory, haptic, olfactory, etc., (iii) Experience Type, and (iv) XR Type as for being VR, AR, or MR and these branch down into their consumption devices.

Second, the Stakeholders consist of (i) Experience in XR which branches down into the type of use and their level of experience, and (ii) Discipline Frameworks that govern the stakeholder's practice.

Finally, the Prototype consists of (i) Selected Modality to be tested in this prototype, (ii) Prototyping Framework which branches into the Approach, and it branches into different characteristics. Furthermore, it branches into the Life Cycle, which branches into different characteristics, (iii) Level of Fidelity, which is the level of realism and simulation accuracy. It has three levels: Low Fidelity (Lo-Fi), Medium Fidelity (Med-Fi), and High Fidelity (Hi-Fi). These branch into Prototype Form, i.e., whether it is Virtual or Digital versus being Physical. Further, it branches down into different methods and techniques, and (iv) Objectives and this branches into, first, the Role or Function and differentiating between them is one of the important discussions around prototyping. There could be intricacies between the two, that cannot be addressed at this stage. For example, a prototype role is its expected behaviour and outcome of the function. This role is different based on the field so in product development its role is to minimise risk and uncertainty and its function is enhancing performance and overall user experience of the final product [2]. A prototype's purpose is the reason for which it is created in order to achieve explicit set objectives. Sometimes, it is also regarded as one of its key characteristics which should be carefully considered when using it. Second, Requirements which branches down into (i) Depth, that refers to the depth of the system functions that are included and it could be classified into Horizontal (flat) and Vertical (multi-layered), and (ii) Scale that refers to the amount of the system functions that are included. Third, Evaluation Purpose which branches into the common 4 types in the literature.

3 CONCLUSION AND FUTURE WORK

In this paper, we synthesized insights from various prototyping taxonomies in the contexts of XR and other disciplines to propose a holistic decision-making guide. We further informed our taxonomy by surveying existing XR prototyping work in the literature. Intended primarily for novices and non-tech-savvy designers, our proposed taxonomy potentially also serves as an educational resource to introduce learners to XR prototyping nuances. We do not claim completion of the work, yet, it is a significant step towards the desired outcome. Next, we validate our hypotheses further through a series of evaluations and user studies. First, we will evaluate its structure, applicability, and purpose as common in related literature. Then, we will conduct user testing in sample scenarios. Future work will focus on validating our hypothesis and updating and automating this decision-making process, similar to the taxonomy automation by Filippi and Barattin [4], leveraging the structure of this taxonomy.

REFERENCES

- [1] N. Ashtari, A. Bunt, J. McGrenere, M. Nebeling, and P. K. Chilana. Creating augmented and virtual reality applications: Current practices, challenges, and opportunities. In *Proc. ACM CHI '20*, p. 1–13, 2020.
- [2] B. Camburn, V. Viswanathan, J. Linsey, D. Anderson, D. Jensen, R. Crawford, K. Otto, and K. Wood. Design prototyping methods: state of the art in strategies, techniques, and guidelines. *Design Science*, 3, 2017.
- [3] L. Colusso, C. L. Bennett, G. Hsieh, and S. A. Munson. Translational resources: Reducing the gap between academic research and hci practice. In *Proc. ACM DIS '17*, p. 957–968, 2017.
- [4] S. Filippi and D. Barattin. Classification and selection of prototyping activities for interaction design. *Intelligent Information Management*, 04:147–156, 2012.
- [5] G. Freitas, M. S. Pinho, M. S. Silveira, and F. Maurer. A systematic review of rapid prototyping tools for augmented reality. In *Proc. Symposium on Virtual and Augmented Reality 2020*, pp. 199–209, 2020.
- [6] M. R. Ogden. The Next Innovation in Immersive [Actuality] Media Isn't Technology- It's Storytelling. In *Proceedings of The Asian Conference on Media, Communication Film 2019*, pp. 97–112. The International Academic Forum (IAFOR), Tokyo, JP, December 2019.
- [7] D. J. Roedl and E. Stolterman. Design research at chi and its applicability to design practice. In *Proc. ACM CHI '13*, p. 1951–1954, 2013.